

Vaishnavi Bommagani

+1 (940)-629-3660 | vbommagani19@gmail.com | [GitHub](#)

PROFESSIONAL EXPERIENCE

Oct 2025 – Present

Software Engineer III, Spectrum, USA

- Designed, developed, documented (Swagger) and delivered **reusable client libraries** using **Java 25, Spring Boot 4** serving as the integration layer between Spectrum and partners such as **Netflix and Apple TV**, reducing onboarding effort.
- Built Python automation scripts and AI-driven utilities** to streamline operational and engineering workflows, including log analysis, entitlement validation, and test data generation; reduced manual effort, improved troubleshooting efficiency, and enabled faster iteration cycles for backend services.
- Improved engineering productivity by 40% by integrating **AI-assisted development workflows** (Kiro, Q), generating boilerplate modules, unit tests, interface contracts, and static-analysis-driven refactoring plans. Built custom prompts for rapid API stub creation, code migration, and complex state-machine transformations, significantly reducing PR cycle time.
- Built and scaled a **reconciliation microservice** and **Kafka-based Command Handler microservices** that consume entitlement events, decompose workflows, and orchestrate calls across libraries, supporting high-throughput event-driven processing.
- Developed and owned **end-to-end entitlement lifecycle features** (Add, Cancel, Change, Suspend, Resume), ensuring accurate and consistent state transitions across partner platforms by **analysing billing data and interpreting billing codes in real time**.
- Ensured **reliable and consistent entitlement processing** across distributed systems by **handling asynchronous activation callback notifications**, applying the **Two-Phase Commit (2PC) pattern** for transactional consistency, and documenting architectural decisions using **Architecture Decision Records (ADRs)**.
- Supported both **synchronous and asynchronous partner integrations**, designing APIs and event-driven flows to accommodate real-time requests as well as non-blocking, message-based processing for high scalability.
- Improved reliability and fault tolerance by implementing **retries, dead-letter topics, and structured exception handling using Resilience4j**, significantly reducing message loss and recurring production incidents.
- Implemented **Spring Security with OAuth 2.0-based authentication and authorization**, ensuring secure access to entitlement APIs and compliance with enterprise security standards.
- Designed and optimized **MongoDB and Document DB collections**, modelling partner data, entitlement data, and transition details, enabling low-latency reads, efficient writes, and scalable entitlement workflows.
- Implemented **unit tests (JUnit, Mockito)** achieving more than **90% code coverage**, along with **integration testing using Cucumber** and end-to-end validation via **Postman and cURL**, improving production reliability.
- Delivered **cloud-native services on AWS (EKS, EC2, MSK, S3)** using **Docker, Jenkins, Kubernetes**, and implemented **CI/CD pipelines with Git-based workflows**, enabling faster, repeatable, and low-risk deployments.
- Improved **operational excellence and service observability** by integrating **Spring Actuator health endpoints**, centralized configuration via **Spring Cloud Config**, and environment-specific configurations across all environments.
- Enforced **high code quality and maintainability standards** through **SonarQube analysis**, ensuring production-ready services.
- Applied established **design patterns and coding standards**, writing **clean, maintainable, and testable code**, and actively participated in **peer code reviews** to improve overall code quality and knowledge sharing.

Full Stack Developer, DataEdge Inc, USA

Jul 2024 – Aug 2025

- Developed full-stack web applications, using **Python, Fast API** for backend and **React.js with Tailwind CSS** for frontend, digitalizing film making workflows and improving development efficiency by **50%**.
- Implemented scalable backend components using Python, **SQLAlchemy ORM** for a pre-production system, enabling efficient script analysis and casting workflow reduced manual planning time by **55%**.
- Built secure Restful APIs using Fast API, enabling seamless communication between frontend and backend services.
- Designed responsive UI for pre-production workflows using React.js, **Web Hooks**, Tailwind CSS, and secure **Axios-REST APIs** communication—improving user task completion rate by **42%**.
- Integrated **LangChain, LangGraph**, and GPT LLMs for AI-based script generation using tuned prompts (temperature: 0.7, max_tokens: 4000), reducing manual writing time by **65%**.
- Created an AI assistant in the documentation module using LangChain memory chains for contextual editing and script suggestions—improving user productivity by **50%**.
- Architected an ETL pipeline using AWS Glue, Lambda, and S3, reducing processing time by **60%**; implemented Redis caching to speed up frequently accessed queries; and deployed scalable solutions via CI/CD with GitHub Actions, Jenkins, Docker, and Kubernetes.
- Developed **Splunk** dashboards and integrated **AWS CloudWatch** with SNS for proactive alerting and real-time monitoring, improving system reliability and reducing mean time to resolution (MTTR) by **35%**.
- Collaborated in sprint planning, PR reviews, and on-call production support in an **Agile** team, maintaining **95% service uptime**.

- Designed and developed backend services using Java, **Spring Boot**, and **Spring MVC** for the “OT2 Central platform”, enabling centralized enterprise management and improving deployment scalability by 50%.
 - Developed subscription configuration module using Java, **Spring MVC**, integrating with internal billing services via **RESTful APIs**, improving onboarding efficiency by 65% and ensuring accurate resource allocation across tenants.
 - Developed of the User Entitlement microservice using **Spring Security**, integrating **OAuth 2.0 and JWT**, reduced unauthorized access attempts by 40% and enhanced role-based access control across multi-tenant systems.
 - Built an event-driven pipeline using **Kafka** to asynchronously process tenant data updates, improving onboarding speed by 45%.
 - Built scalable **RESTful APIs** for tenant registration, enabling seamless multi-tenant onboarding and reducing manual admin effort by 60%. Integrated APIs with **Angular** components and documented using **Swagger UI**, accelerating downstream integration by 40%.
 - Optimized **PostgreSQL** queries using **Spring Data JPA** and **Hibernate ORM**, enhancing database performance under heavy load by 40%, and ensuring transactional integrity for critical user entitlement data.
 - Automated CI/CD workflows for microservices using **Git**, **Jenkins**, **Docker**, and **JUnit**, achieving 90% integration and unit test coverage, ensuring seamless zero-downtime deployments.
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TECHNICAL SKILLS

- **Programming Languages:** Java, Python, Object Oriented Programming, C, C++, PL/SQL
 - **Web Technologies & DB:** React.js, Vue.js, Angular, Node.js, MySQL, NoSQL, MongoDB, JavaScript, TypeScript, HTML, CSS, D3.js
 - **Frameworks and API:** Spring Boot, Spring MVC, REST APIs, Microservices, Hibernate, Maven, Fast APIs, SQLAlchemy.
 - **Cloud and Proficient tools:** AWS (DynamoDB, Lambda, EC2, SQS, S3, MSK), SCRUM oriented Agile, Design Patterns, Test Driven Development (TDD), JUnit, Mockito, Kubernetes, Jenkins, CI/CD, Docker, Git, WEKA, Azure
 - **AI/ML Frameworks & Libraries:** LangChain, LangGraph, Scikit-learn, PyTorch, TensorFlow, Regression Models, Traditional ML algorithms, Prompt Engineering, LLM-based workflows, Pandas, Numpy, EDA, Data Cleaning
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ACADPROJECTS

RAG based ChatBot:

- RAG-based chatbot using **Gemini and Text-Bison LLMs** with FAISS and Pinecone for Wikipedia-based semantic search; achieved 85%+ accuracy and improved contextual response quality via a Gradio interface.
- Designed an **end-to-end retrieval and re-ranking pipeline**, optimizing chunking, embeddings, and prompt engineering to reduce hallucinations and improve response relevance and latency.

Leave Manager:

- Developed a Leave Management MCP Server using FastMCP (Model Context Protocol) in Python, enabling AI tools for leave balance checks, leave applications with specific dates, and leave history retrieval. Used UV Python package manager for fast, reliable dependency management, showcasing clean backend architecture and AI-tool integration.

Smart SQL Assistant:

- A natural language-to-SQL tool using **Fast API**, **LangChain**, **LangGraph**, and **PostgreSQL** to help users query data using plain English. Designed AI agents for generating SQL queries, reducing the need for manual SQL writing by **50%**.
- Implemented **query validation**, **schema-aware prompting**, and **execution safeguards** to ensure syntactically correct, secure, and performant SQL generation across complex database schemas.

Neighbourly:

- Developed a web platform enabling community feedback and complaint-sharing to improve neighborhood issues while protecting user privacy. Developed using **Angular**, **Spring boot**, **PostgreSQL**, **Rest APIs**.
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EDUCATION

Master of Science, Computer Science

University of North Texas

Denton, Texas

Bachelor of Technology, Computer Science and Engineering

CVR College of Engineering

Hyderabad, India